Lines & Angles

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Deductive reasoning**

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 5(5) + 4 = 29$.

2 **[Foundational]** • **Target: Deductive reasoning**

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Final Answer: $x = 4$ and $x = 6$. Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

3 **[Foundational]** • **Target: Properties of equality and congruence**

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 96°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 45^\circ) = 96^\circ$.

4 **[Foundational]** • **Target: Properties of equality and congruence**

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

5 **[Foundational]** • **Target: Properties of equality and congruence**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 8. Substituting $x = 1$ into $y = 3x + 5$ gives $y = 3(1) + 5 = 8$.

6 **[Foundational]** • **Target: Properties of equality and congruence**

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 89°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 52^\circ) = 89^\circ$.

7 **[Foundational]** • **Target: Properties of equality and congruence**

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Properties of equality and congruence**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 11. Substituting $x = 2$ into $y = 3x + 5$ gives $y = 3(2) + 5 = 11$.

9 **[Intermediate]** • **Target: Properties of equality and congruence**

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 82°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 59^\circ) = 82^\circ$.

10 **[Intermediate]** • **Target: Properties of equality and congruence**

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 32 cm. Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

11 **[Intermediate]** • **Target: Properties of equality and congruence**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 14. Substituting $x = 3$ into $y = 3x + 5$ gives $y = 3(3) + 5 = 14$.

12 **[Intermediate]** • **Target: Properties of equality and congruence**

In ΔABC studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 75°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 66^\circ) = 75^\circ$.

13 **[Intermediate]** • **Target: Properties of equality and congruence**

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 40 cm. Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

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14 [Intermediate] • Target: Properties of equality and congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 17. Substituting $x = 4$ into $y = 3x + 5$ gives $y = 3(4) + 5 = 17$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 7.01 Circles

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(2)$.

Final Answer: 10. Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

16 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 8.01 Area, Surface Area & Volume

A data set recorded for proving segment relationships consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

17 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 9.01 Transformations & Symmetry

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(3)$.

Final Answer: 12. Substitute $x = 3$: $f(3) = 2(3) + 6 = 12$.

18 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 1.01 Foundations & Reasoning

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(4)$.

Final Answer: 14. Substitute $x = 4$: $f(4) = 2(4) + 6 = 14$.

19 [Challenging] • Target: Proving segment relationships • Cross-Lesson: 10.01 Coordinate Geometry

For the model $f(x) = 2x + 6$ in proving segment relationships, evaluate $f(5)$.

Final Answer: 16. Substitute $x = 5$: $f(5) = 2(5) + 6 = 16$.

20 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 11.01 Probability

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 94°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 45^\circ) = 94^\circ$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Conditional statements

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

22 [Foundational] • Target: Conditional statements

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

23 [Foundational] • Target: Conditional statements

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(5)$.

a. 24 ✓

b. 27

c. 15

d. 28

Substitute $x = 5$: $f(5) = 3(5) + 9 = 24$.

24 [Foundational] • Target: Language of logic

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(2)$.

a. 11 ✓

b. 15

c. 8

d. 15

Substitute $x = 2$: $f(2) = 4(2) + 3 = 11$.

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25 [Foundational] • Target: Language of logic

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

26 [Foundational] • Target: Language of logic

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(5)$.

a. 23 ✓

b. 27

c. 20

d. 27

Substitute $x = 5$: $f(5) = 4(5) + 3 = 23$.

27 [Foundational] • Target: Deductive reasoning

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(2)$.

a. 14 ✓

b. 19

c. 10

d. 18

Substitute $x = 2$: $f(2) = 5(2) + 4 = 14$.

28 [Foundational] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Conditional statements

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

30 [Intermediate] • Target: Conditional statements

For the model $f(x) = 3x + 9$ in conditional statements, evaluate $f(3)$.

a. 18 ✓

b. 21

c. 9

d. 22

Substitute $x = 3$: $f(3) = 3(3) + 9 = 18$.

31 [Intermediate] • Target: Conditional statements

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

32 [Intermediate] • Target: Language of logic

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

33 [Intermediate] • Target: Language of logic

For the model $f(x) = 4x + 3$ in language of logic, evaluate $f(3)$.

a. 15 ✓

b. 19

c. 12

d. 19

Substitute $x = 3$: $f(3) = 4(3) + 3 = 15$.

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34 [Intermediate] • Target: Language of logic

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

35 [Intermediate] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

36 [Intermediate] • Target: Deductive reasoning

For the model $f(x) = 5x + 4$ in deductive reasoning, evaluate $f(3)$.

a. 19 ✓

b. 24

c. 15

d. 23

Substitute $x = 3$: $f(3) = 5(3) + 4 = 19$.

37 [Intermediate] • Target: Deductive reasoning

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 10

c. 8 ✓

d. 11

Substituting $x = 1$ into $y = 3x + 5$ gives $y = 3(1) + 5 = 8$.

39 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 10.01 Coordinate Geometry

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 8

b. 13

c. 11 ✓

d. 14

Substituting $x = 2$ into $y = 3x + 5$ gives $y = 3(2) + 5 = 11$.

40 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

41 [Challenging] • Target: Properties of equality and congruence • Cross-Lesson: 3.01 Triangles & Congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 14

b. 19

c. 17 ✓

d. 20

Substituting $x = 4$ into $y = 3x + 5$ gives $y = 3(4) + 5 = 17$.

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42 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

43 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 52^\circ) = 87^\circ$.

44 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 80° ✓

b. 90°

c. 72°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 59^\circ) = 80^\circ$.

45 [Challenging] • Target: Proving angle relationships • Cross-Lesson: 7.01 Circles

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

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